(70) (or)

By using units method.

Total work in units = LCM(14, 21) = 42 units

Eff of A → $\frac{42}{14}$ = 3 units, Eff of B = $\frac{42}{21}$ = 2 units

Combined efficiency = 3+2 = 5 units.

Last 3 day B only worked, means he completes

3 days × 2 units on each day = 6 units of work.

Remaining work = total − 6 units = 42 − 6 = 36 units

Remaining work done by both A & B. in

5 units — 1 day] together ⇒ $\frac{36 \times 1}{5} = 7\frac{1}{5}$ days

36 units — ?

Total days workdone = $7\frac{1}{5} + 3 = 10\frac{1}{5}$ days.

Ex:-15:- [B.Q:-11] :- A group of men completes a work in 10 days, but 5 of them absent. So, the rest men completed it in 12 days. Original no. of Men?

$\frac{W_1}{M_1 D_1} = \frac{W_2}{M_2 D_2}$ [here work constant.]

[M_1 → Original → as 'x'
M_2 → 5 less to M_1 → x−5]

$M_1 D_1 = M_2 D_2$

$x \times 10 = (x-5)12 \Rightarrow 10x = 12x - 60$

$\Rightarrow 12x - 10x = 60 \Rightarrow$ **x = 30 Men**

Ex:-16:- [B.Q:-14] :- A = $\frac{1}{15}$, B = $\frac{1}{10}$, B started and left after 5 days. In how many days that A takes to complete the remaining work?

$mA + nB = 1$

lets assume x days by A; B days → 5

$\frac{x}{15} + \frac{5}{10} = 1 \Rightarrow \frac{2x+15}{30} = 1.$

$\Rightarrow$ **x = 7.5 days**

Ex:-17:- [B.Q:16] :- $A = \frac{1}{80}$, A worked for 10 days, remaining work done by B in 42 days. Together in how many days?

Ans: Let's assume B can take x days to complete.

$$10 \cdot \frac{1}{80} + 42 \cdot \frac{1}{x} = 1 \Rightarrow \frac{10x + 80 \times 42}{80x} = 1$$

$$80x - 10x = 80 \times 42 \Rightarrow 70x = 80 \times 42$$

$$x = 48 \text{ days.}$$

$$\text{together} = \frac{80 \times 48}{80 + 48}$$

$$= \frac{80 \times 48}{128} = 30 \text{ days}$$

Ex:-18:- [B.Q:-19] :- A does half as much work as B in $\frac{3}{4}$th of the time. If together they take 18 days to complete the work, how much time shall B take to do it?

Ans:- Now days by A → x days :] Assume.
" " B → y days

A does $\frac{1}{2}$ work in $\frac{x}{2}$ days, B does $\frac{3}{4}$th of work in $\frac{3}{4}$th time means $\frac{3}{4}y$.

$$\frac{x}{2} = \frac{3}{4}y \Rightarrow \frac{x}{y} = \frac{6}{4} = \frac{3}{2}$$

$x : y = 3 : 2$ [time / Total no of days.]

$$\text{together} = \frac{3a \times 2a}{3a + 2a} = 18 \Rightarrow \frac{6a \times 6a}{5a} = 18 \Rightarrow a = 15 \text{ days}$$

B time = $2a = 2 \times 15 = 30$ days.

A time = $3a = 3 \times 15 = 45$ days.